A FIELD PROJECT REPORT ON

SMART AGRICULTURE MONITORING AND AUTOMATION SYSTEM

Submitted

in partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

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CERTIFICATE

This is to certify that the field project entitled entitled **"A field project report on IOT- based smart agriculture monitoring and automatic system"** is being submitted by **(Ch Deepika & 241FA04308), (O Yamini &241FA04858),(G Rahul & 241FA04D59),(G Manolini &241FA04D89)** in partial fulfilment of the requirements for the degree of Bachelor of Technology (B.Tech) in **Computer Science and Engineering** at Vignan's Foundation For Science Technology & Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India.

This is a bona fide work carried out by the aforementioned students under my guidance and supervision.

Guide

Project Review Committee **Head of the Department**

DECLARATION

Date:

We hereby declare that the work presented in the field project titled **"A field project report on IOT- based smart agriculture monitoring and automatic system"** is the result of our own efforts and investigation. This project is being submitted under the supervision of **(Mr.Kumar Devapogu, Asst.Prof, CSE)**, in partial fulfilment of the requirements for the Bachelor of Technology (B.Tech) degree in Computer Science and Engineering at Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh.

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A FIELD PROJECT REPORT ON

IoT-BASED SMART AGRICULTURE MONITORING AND AUTOMATION SYSTEM

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CHAPTER 1

INTRODUCTION

Agriculture is the backbone of many developing countries, yet traditional farming methods often suffer from inefficient resource usage and lack of real-time monitoring. With the rapid advancement of Internet of Things (IoT) technology, agriculture can be transformed into a smarter and more efficient system.

The **IoT-Based Smart Agriculture Monitoring and Automation System** aims to help farmers monitor soil moisture, temperature, humidity, and other environmental parameters in real time. Using sensors, microcontrollers, and cloud platforms, the system automates irrigation and provides accurate data to farmers through mobile or web applications. This reduces manual effort, conserves water, and improves crop productivity.

1.1 Problem Definition

Traditional agriculture faces several challenges such as:

- Excessive or insufficient irrigation
- Lack of real-time soil and climate monitoring
- Manual labor dependency
- Water wastage and low crop yield

Farmers often rely on guesswork rather than data, which leads to inefficient farming practices. There is a need for a smart system that can monitor agricultural conditions continuously and automate farming operations.

1.2 Existing System

- Manual monitoring of soil and weather conditions
- Fixed irrigation schedules regardless of soil moisture
- High labor cost and time consumption
- No real-time alerts or remote access
- Inefficient water and energy usage

1.3 Proposed System

- Uses IoT sensors to monitor soil moisture, temperature, and humidity
- Automated irrigation based on real-time sensor data
- Cloud integration for data storage and analysis
- Mobile/web application for remote monitoring
- Reduces water wastage and improves crop yield
- Provides alerts and reports to farmers

1.4 Literature Review

Recent studies show that IoT-based agricultural systems significantly improve productivity and resource management. Many researchers have implemented smart irrigation using soil moisture sensors and microcontrollers like Arduino and ESP8266. Cloud platforms enable real-time monitoring and predictive analysis, helping farmers make data-driven decisions. However, most systems lack full automation and user-friendly interfaces, which this project aims to address.

CHAPTER 2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

Hardware Requirements:

- Microcontroller: Arduino / ESP8266 / ESP32
- Soil Moisture Sensor
- Temperature & Humidity Sensor (DHT11/DHT22)
- Water Pump / Relay Module
- Power Supply
- Internet Connectivity

Software Requirements:

- Operating System: Windows / Linux
- Programming Language: Embedded C / Python
- IoT Platform: ThingSpeak / Firebase / Blynk
- IDE: Arduino IDE
- Web/Mobile App (optional): HTML, CSS, JavaScript

2.2 Software Requirements Specification (SRS)

Functional Requirements (FR):

- Read sensor data continuously
- Display real-time data on dashboard
- Automatic irrigation control
- Manual override option
- Alert notifications

Non-Functional Requirements (NFR):

- Reliable and accurate data collection
- Low power consumption
- Easy to use interface
- Secure data transmission
- High system availability

CHAPTER 3

SYSTEM DESIGN

3.1 Modules of the System

1. **Sensor Module**
   - Collects soil moisture, temperature, and humidity data
2. **Control Module**
   - Controls irrigation using relay and water pump
3. **IoT & Cloud Module**
   - Sends data to cloud platform for storage and analysis
4. **User Interface Module**
   - Displays data on mobile/web application
5. **Alert Module**
   - Sends notifications when conditions cross threshold levels

3.2 UML Diagrams

- **Use Case Diagram:** Shows interaction between farmer and system
- **Class Diagram:** Describes sensor, controller, cloud, and user classes
- **Activity Diagram:** Flow from sensing → decision → irrigation
- **Sequence Diagram:** Data flow between sensors, controller, cloud, and user

CHAPTER 4

IMPLEMENTATION

4.1 Sample Code (Arduino – Soil Moisture Based Irrigation)

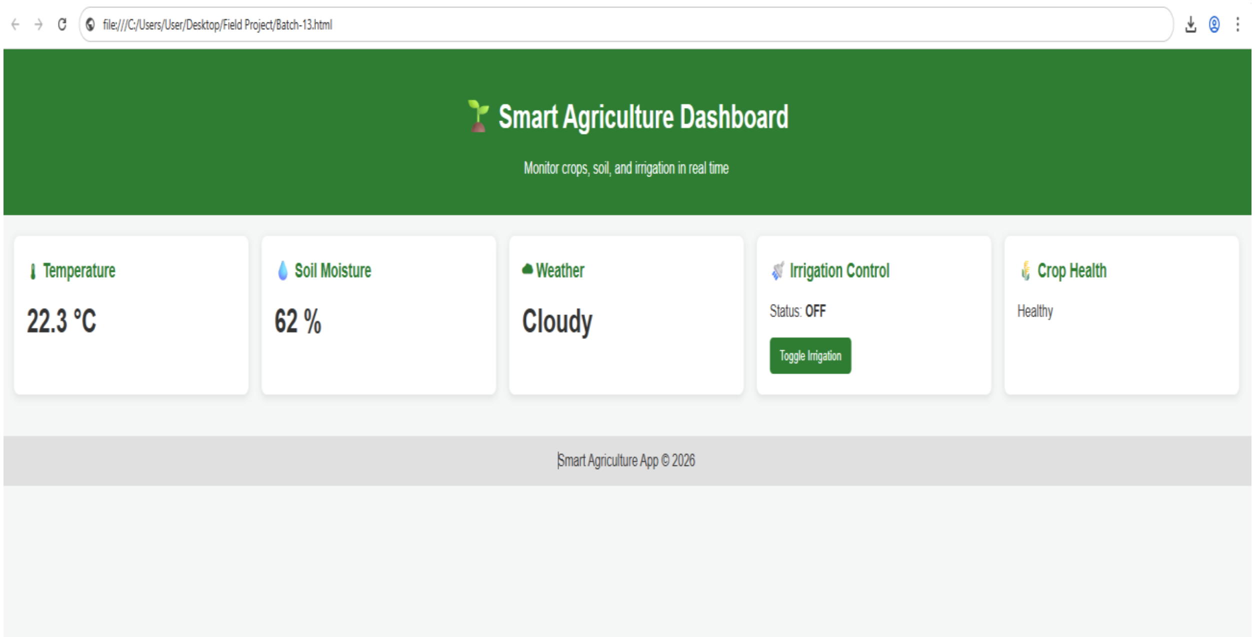
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Smart Agriculture App</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f4f7f6;
      color: #333;
    }

    header {
      background: #2e7d32;
      color: white;
      padding: 20px;
      text-align: center;
    }

    .container {
      padding: 20px;
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
      gap: 20px;
    }

    .card {
      background: white;
      border-radius: 8px;
      padding: 20px;
      box-shadow: 0 2px 8px rgba(0,0,0,0.1);
```

```
        }

        .card h3 {
            margin-top: 0;
            color: #2e7d32;
        }

        .value {
            font-size: 2em;
            font-weight: bold;
        }

        button {
            background: #2e7d32;
            color: white;
            border: none;
            padding: 10px 15px;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background: #1b5e20;
        }

        footer {
            text-align: center;
            padding: 15px;
            background: #e0e0e0;
            margin-top: 20px;
        }
    </style>
</head>

<body>

<header>
    <h1>🌱 Smart Agriculture Dashboard</h1>
    <p>Monitor crops, soil, and irrigation in real time</p>
</header>
```

```html
<div class="container">

  <div class="card">
    <h3>🌡 Temperature</h3>
    <div class="value" id="temp">-- °C</div>
  </div>

  <div class="card">
    <h3>💧 Soil Moisture</h3>
    <div class="value" id="moisture">-- %</div>
  </div>

  <div class="card">
    <h3>☁ Weather</h3>
    <div class="value" id="weather">Sunny</div>
  </div>

  <div class="card">
    <h3>🚿 Irrigation Control</h3>
    <p>Status: <strong id="irrigationStatus">OFF</strong></p>
    <button onclick="toggleIrrigation()">Toggle Irrigation</button>
  </div>

  <div class="card">
    <h3>🌾 Crop Health</h3>
    <p id="cropStatus">Healthy</p>
  </div>

</div>

<footer>
  Smart Agriculture App © 2026
</footer>

<script>
  function generateRandomData() {
    document.getElementById("temp").innerText =
      (20 + Math.random() * 10).toFixed(1) + " °C";
```

```
        document.getElementById("moisture").innerText =
            (40 + Math.random() * 30).toFixed(0) + " %";

        const weatherOptions = ["Sunny", "Cloudy", "Rainy"];
        document.getElementById("weather").innerText =
            weatherOptions[Math.floor(Math.random() * weatherOptions.length)];

        const moisture = parseInt(document.getElementById("moisture").innerText);
        document.getElementById("cropStatus").innerText =
            moisture < 50 ? "Needs Water" : "Healthy";
    }

    function toggleIrrigation() {
        const status = document.getElementById("irrigationStatus");
        if (status.innerText === "OFF") {
            status.innerText = "ON";
            status.style.color = "green";
        } else {
            status.innerText = "OFF";
            status.style.color = "red";
        }
    }

    // Update sensor data every 5 seconds
    setInterval(generateRandomData, 5000);
    generateRandomData();
</script>

</body>
</html>
```

Out put:



4.2 Test Cases

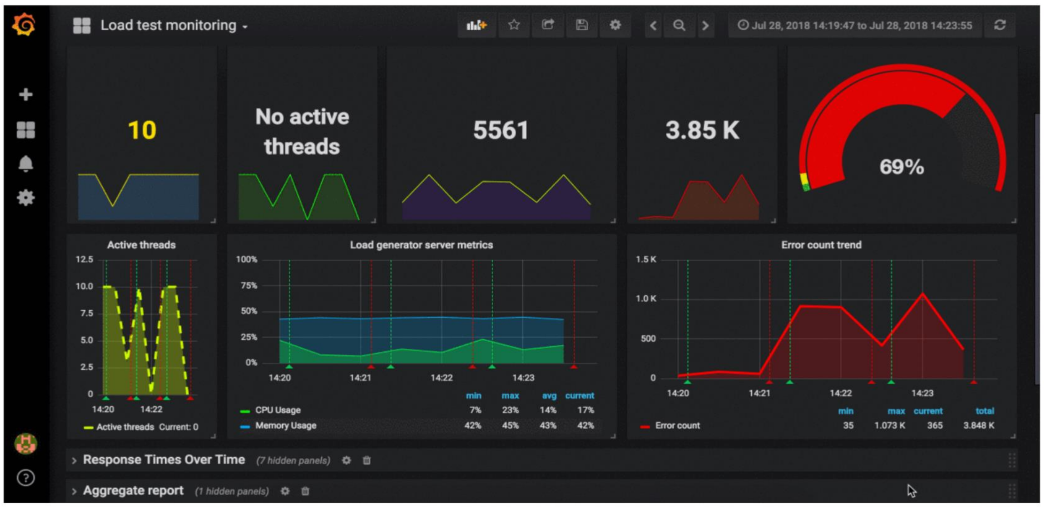
Test Case	Input Condition	Expected Output
TC01	Dry soil	Pump ON
TC02	Wet soil	Pump OFF
TC03	Sensor failure	Alert generated
TC04	Internet loss	Local control active

CHAPTER 5

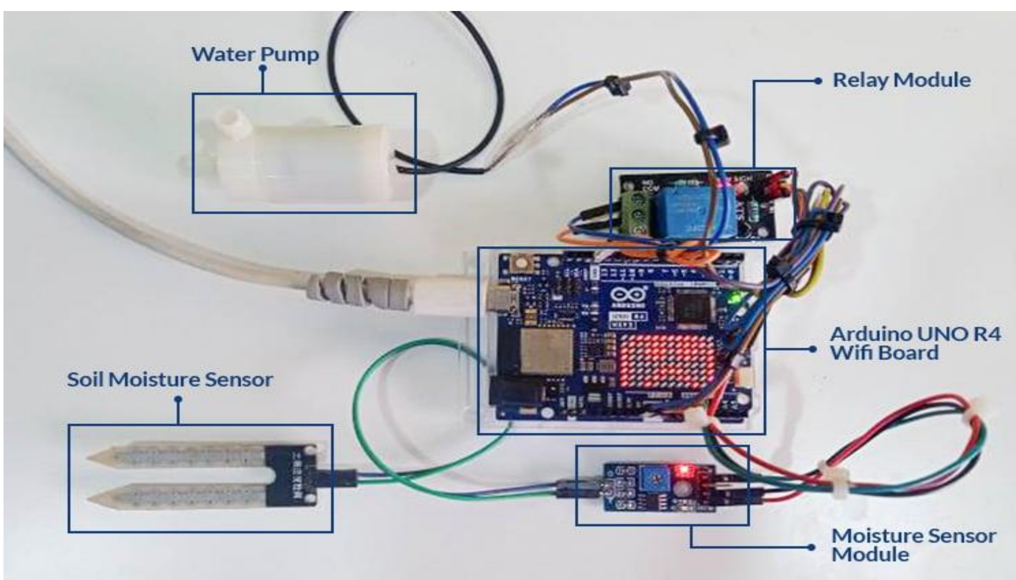
RESULTS

5.1 Output Screens

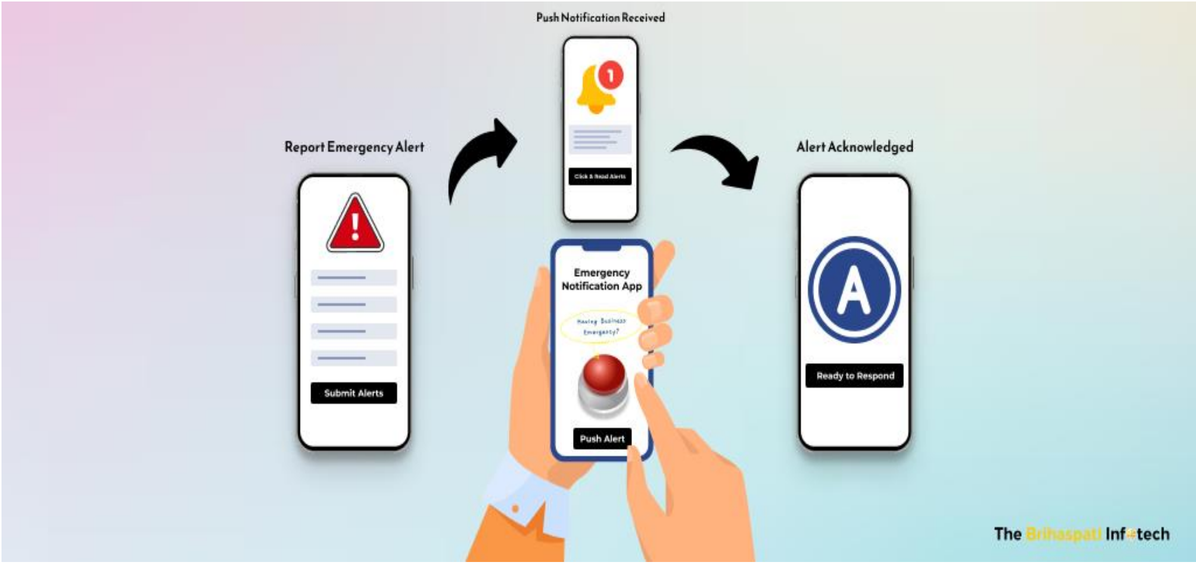
- **REAL-TIME SENSOR DATA DISPLAYED ON DASHBOARD**



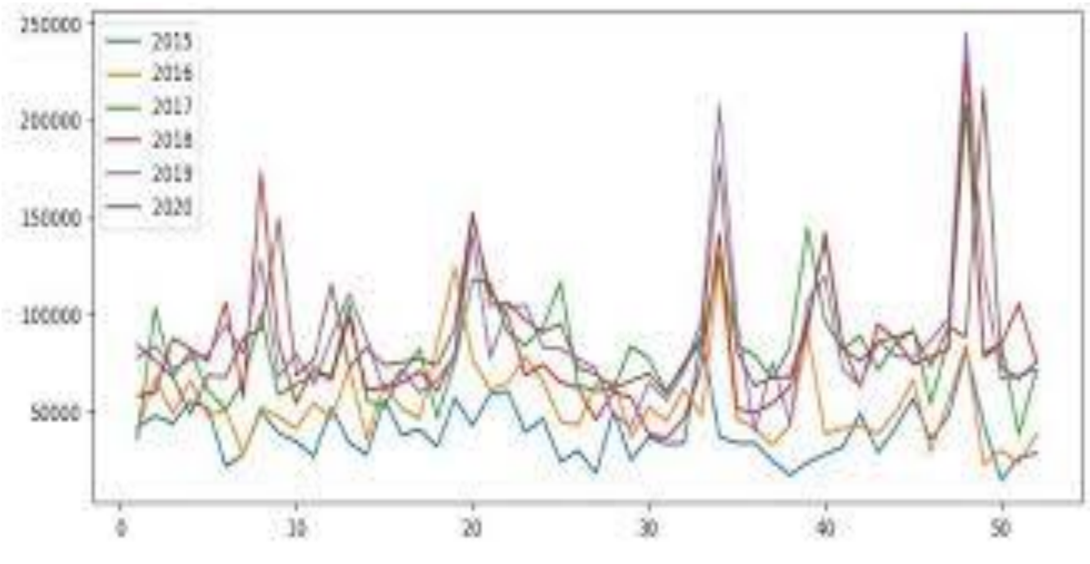
z5.2AUTPUMP ON/OFF BASED ON SOIL MOIST



5.3 ALERTS SENT TO USER APPLICATION



5.4 HISTORICAL DATA GRAPHS FOR ANALYSIS



The system successfully reduced water usage and improved irrigation efficiency.

CHAPTER 6

CONCLUSION

The **IoT-Based Smart Agriculture Monitoring and Automation System** provides an efficient and reliable solution for modern farming. By automating irrigation and enabling real-time monitoring, the system minimizes water wastage, reduces labor dependency, and enhances crop productivity. This project demonstrates how IoT technology can revolutionize agriculture and support sustainable farming practices.

PROJECT LINK: https://github.com/manolinig197-del/SmartAgriculture

CHAPTER 7

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